

Apache Paimon: Lakehouse Storage for Real-Time Analytics

Apache Paimon brings a streaming-first approach to lakehouse storage, making it ideal for real-time analytics and evolving data workloads.



Modern data platforms increasingly demand real-time analytics, where insights are generated as data arrives rather than hours later. Traditional data lakes work well for batch processing but often struggle with low-latency updates, frequent data changes, and incremental processing.

Streaming systems, on the other hand, are optimised for real-time ingestion but lack robust storage semantics such as schema evolution, time travel, and efficient updates. This gap has led to the rise of lakehouse architectures, combining the reliability of data lakes with the performance of streaming systems.

Apache Paimon addresses this challenge by providing an open source table storage format designed specifically for streaming-first, real-time analytics.

Overview of Apache Paimon

Apache Paimon is an open source lake house storage

framework built for continuous data ingestion, upserts, and incremental processing. It uses LSM-tree (Log-Structured Merge Tree) principles, like modern databases, to efficiently merge small updates into larger files.

Unlike traditional append-only table formats, Paimon is designed around primary-key tables, making it suitable for:

- Change Data Capture (CDC)
 - Streaming updates
 - Near real-time analytics
- Apache Paimon sits between:
- Streaming engines (for ingestion and updates)
 - Storage systems (HDFS or object storage)
 - Analytics and ML workloads

It is most used with Apache Flink, enabling unified streaming and batch workflows.

Apache Paimon architecture

At a high level, Apache Paimon consists of three core layers: